

Antibody-Drug Conjugates (ADC) in HER2/neu-Positive Gynecologic Tumors.

PMID: 37959808 | Indexed in PubMed

Antibody-drug conjugates (ADCs) are a new class of targeted anti-cancer therapies that combine a monoclonal tumor-surface-receptor-targeting antibody with a highly cytotoxic molecule payload bonded through specifically designed cleavable or non-cleavable chemical linkers. One such tumor surface receptor is human epidermal growth factor 2 (HER2), which is of interest for the treatment of many gynecologic tumors. ADCs enable the targeted delivery of a variety of cytotoxic therapies to tumor cells while minimizing delivery to healthy tissues. This review summarizes the existing literature about HER2-targeting ADC therapies approved for use in gynecologic malignancies, relevant preclinical studies, strategies to address ADC resistance, and ongoing clinical trials.

Use of Antibody-Drug Conjugates in the Early Setting of Breast Cancer.

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Antibody-drug conjugates (ADCs) are anticancer agents with the capacity to selectively deliver their payloads to cancer cells. Antibody-drug conjugates consist of a monoclonal antibody backbone connected by a linker to cytotoxic payloads. Antibody-drug conjugate effect occurs either by directly targeting cancer cells via membrane antigen or through "bystander effect." Antibody-drug conjugates have demonstrated efficacy against various types of tumors, including breast cancer. Ado-trastuzumab emtansine is presently the only approved ADC for the treatment of breast cancer in the early setting, while several ADCs are now approved for metastatic breast cancer. Due to the transformative impact that several ADCs have reported in the setting of advanced breast cancer, researchers are now testing more of such compounds in the early setting, to portend benefits to patients through highly potent anticancer drugs. Ongoing trials hold the potential to transform treatment protocols for early breast cancer in the near future. These trials are aiming at evaluating different treatment modulation approaches, as informed by breast cancer risk of recurrence, including toward treatment de-escalation. Efforts are provided in ongoing clinical trials to identify the patients who will benefit most, to pursue paradigms of precision medicine with the novel ADCs. This review focuses on the potential role of ADCs in early breast cancer, providing an overview of the latest progress in their development and how they are implemented in ongoing clinical trials.

Antibody-Drug Conjugates: A promising breakthrough in cancer therapy.

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Antibody-drug conjugates (ADCs) provide effective cancer treatment through the selective delivery of cytotoxic payloads to the cancer cells. They offer unparalleled precision and specificity in directing drugs to cancer cells while minimizing off-target effects. Despite several advantages, there is a requirement for innovations in the molecular design of ADC owing to drug resistance, cancer heterogeneity along the adverse effects of treatment. The review critically analyses ADC function mechanisms, unraveling the intricate interplay between antibodies, linkers, and payloads in facilitating targeted drug delivery to cancer cells. The article also highlights notable advancements in antibody engineering, which aid in creating highly selective and potent ADCs. Additionally, the review details significant progress in clinical

ADC development with an in-depth examination of pivotal trials and approved formulations. Antibody Drug Conjugates (ADCs) are a ground-breaking approach to targeted drug delivery, especially in cancer treatment. They offer unparalleled precision and specificity in directing drugs to cancer cells while minimizing off-target effects. This review provides a comprehensive examination of the current state of ADC development, covering their design, mechanisms of action, and clinical applications. The article emphasizes the need for greater precision in drug delivery and explains why ADCs are necessary.

Mechanisms of Resistance to Antibody-Drug Conjugates.

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Antibody-drug conjugates (ADCs), with antibodies targeted against specific antigens linked to cytotoxic payloads, offer the opportunity for a more specific delivery of chemotherapy and other bioactive payloads to minimize side effects. First approved in the setting of HER2+ breast cancer, more recent ADCs have been developed for triple-negative breast cancer (TNBC) and, most recently, hormone receptor-positive (HR+) breast cancer. While antibody-drug conjugates have compared favorably against traditional chemotherapy in some settings, patients eventually progress on these therapies and require a change in treatment. Mechanisms to explain the resistance to ADCs are highly sought after, in hopes of developing next-line treatment options and expanding the therapeutic windows of existing therapies. These resistance mechanisms are categorized as follows: change in antigen expression, change in ADC processing and resistance, and efflux of the ADC payload. This paper reviews the recently published literature on these mechanisms as well as potential options to overcome these barriers.

DNA damaging agent-based antibody-drug conjugates for cancer therapy.

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Currently, four antibody-drug conjugates (ADCs) are approved by the Food and Drug Administration or the European Medicine Agency to treat cancer patients. More than 60 ADCs are in clinical development for cancer therapy. More than 60% of ADCs in clinical trials employ microtubule inhibitors as their payloads. A better understanding of payloads other than microtubule inhibitors, especially DNA-damaging agents, is important for further development of ADCs. In this review, we highlight an emerging trend of using DNA-damaging agents as payloads for ADCs. This review summarizes recent advances in our understanding gained from ongoing clinical studies; it will help to define the utility of DNA-damaging payloads for ADCs as cancer therapeutics. Future directions of the development of ADCs are also discussed, focusing on targeting drug resistance and combination treatment with immunotherapy.

Antibody-drug conjugates for lymphoma patients: preclinical and clinical evidences.

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Antibody-drug conjugates (ADCs) are a recent, revolutionary approach for malignancies treatment, designed to provide superior efficacy and specific targeting of tumor cells, compared to systemic cytotoxic chemotherapy. Their structure combines highly potent anti-cancer drugs (payloads or

warheads) and monoclonal antibodies (Abs), specific for a tumor-associated antigen, via a chemical linker. Because the sensitive targeting capabilities of monoclonal Abs allow the direct delivery of cytotoxic payloads to tumor cells, these agents leave healthy cells unharmed, reducing toxicity. Different ADCs have been approved by the US Food and Drug Administration (FDA) and the European Medicines Agency (EMA) for the treatment of a wide range of malignant conditions, both as monotherapy and in combination with chemotherapy, including for lymphoma patients. Over 100 ADCs are under preclinical and clinical investigation worldwide. This paper provides an overview of approved and promising ADCs in clinical development for the treatment of lymphoma. Each component of the ADC design, their mechanism of action, and the highlights of their clinical development progress are discussed.

[Basics and Innovations in Antibody-Drug Conjugates].

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The efficacy of antibody-drug conjugates (ADCs) has been well-established in clinical use, prompting ongoing research and development efforts to advance the field of ADCs further. This chapter provides overview of the structure and mechanism of action of ADCs, detailing the roles of its components: the antibody, the payload, and the linker. ADCs leak from cancer vessels to bind to specific antigens on the surface of tumor cells. Upon binding and internalization, the drug is released by metabolic enzymes, such as peptidases in the lysosome. The therapeutic window of the conjugated drug is therefore expanded. To enhance the efficacy of ADCs, the chapter will also explore the development of methodologies for generating homogeneous ADCs and the concept of bystander effects, which are particularly relevant in the tumor microenvironment. Additionally, innovative approaches such as radioimmunotherapy, utilizing alpha-ray emitting radionuclides, and photoimmunotherapy, are discussed as promising next-generation ADC strategies. The integration of ADCs with immunotherapy may offer amplified effectiveness through synergistic actions. The chapter underscores that the prolonged therapeutic impact of ADCs cannot be solely attributed to the targeted delivery and controlled release of the payload. A comprehensive understanding from the perspective of cancer immunology is imperative for elucidating the underlying mechanisms contributing to their sustained efficacy.

Antibody-Drug Conjugate-Based Therapeutics: State of the Science.

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Antibody-drug conjugates (ADCs) are complex engineered therapeutics consisting of monoclonal antibodies, directed toward tumor-associated antigens, to which highly potent cytotoxic agents are attached using chemical linkers. This targeted drug delivery strategy couples the precision of the antibody targeting moiety with the cytotoxic activity of the payload, which is generally too toxic on its own to be systemically administered. In this manner, ADCs confer a means to reduce off-target toxicities in patients by limiting the exposure of normal tissues to the payload, thus broadening the potential therapeutic window compared with traditional chemotherapy. The pace of ADC development is accelerating, with the number of investigational agents in human trials having more than tripled over the past 5 years, underscoring the enthusiasm for this transformative approach to cancer treatment. Here, we review the key structural elements of ADC design (antibody, linker, and payload), highlighting critical aspects and technological advances that have affected the clinical effectiveness of this class of biopharmaceuticals. The ADC field continues to evolve, including ongoing efforts aimed at improving

target selection, developing payloads with varied mechanisms of action and increased potency, designing innovative bioconjugation strategies, as well as maximizing efficacy and tolerability in patients. An overview of the current clinical trial landscape is provided, with emphasis on the clinical experience of the four ADCs to have received regulatory approval to date, as well as additional promising candidates currently in late-stage clinical development in both solid tumor and hematological malignancies.

Containment challenges in HPAPI manufacture for ADC generation.

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Antibody-drug conjugates (ADCs) are emerging as an impactful class of therapeutics for the treatment of cancer because of their ability to harness the specificity of an antibody and the cytotoxic potential of the payload to target and destroy cancer cells. However, the potent nature of the cytotoxic payload creates associated manufacturing challenges for active pharmaceutical ingredient (API) manufacturers, because huge investment in containment technology is required to ensure the protection of operators and the environment. Here, we examine the differing attitudes to high-potency categorisation and levels of containment control. We also provide an overview of the most widely used containment strategies for facility design, powder handling, purification, analysis, and cleaning. Finally, we briefly consider the health and safety regulatory challenges associated with the manufacture of cytotoxic payloads for use in antibody-drug conjugates.

Antibody Drug Conjugates: Application of Quantitative Pharmacology in Modality Design and Target Selection.

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Antibody drug conjugates (ADCs) are a multi-component modality comprising of an antibody targeting a cell-specific antigen, a potent drug/payload, and a linker that can be processed within cellular compartments to release payload upon internalization. Numerous ADCs are being evaluated in both research and clinical settings within the academic and pharmaceutical industry due to their ability to selectively deliver potent payloads. Hence, there is a clear need to incorporate quantitative approaches during early stages of drug development for effective modality design and target selection. In this review, we describe a quantitative approach and framework for evaluation of the interplay between drug- and systems-dependent properties (i.e., target expression, density, localization, turnover, and affinity) in order to deliver a sufficient amount of a potent payload into the relevant target cells. As discussed, theoretical approaches with particular considerations given to various key properties for the target and modality suggest that delivery of the payload into particular effect cells to be more sensitive to antigen concentrations for targets with slow turnover rates as compared to those with faster internalization rates. Further assessments also suggest that increasing doses beyond the threshold of the target capacity (a function of target internalization and expression) may not impact the maximum amount of payload delivered to the intended effect cells. This article will explore the important application of quantitative sciences in selection of the target and design of ADC modalities.